atomQUEST 2026 · Round 2
Team: The Collector · Alliance University, Bengaluru
Track 2 — Water purifier (confirm track number)
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The Collector — progress report

An instrumented domestic RO purifier that measures the water path: per-stage pressure drop, pump health, and a fail-locked dispense.

1 TEAM, TRACK AND AREA OF EXPLORATION

The Collector — Alliance University, Bengaluru. **Water purifier track (Track 2)**, area of exploration unchanged from the abstract: **measured filter health in domestic RO purifiers**. Team of two: Prahalad Singh Gaur (sensors, firmware, validation) and Sanskar Kharya (mechanical integration, enclosure, test rig).

2 THE PROBLEM, AND HOW WE ARE SOLVING IT

Filter life in a domestic RO is **estimated, never measured** — days and litres converted to a percentage. The only real sensors, TDS in and out, cannot see the water path: TDS reads conductivity, so it cannot detect a housing refitted **without its cartridge**, a bypassed membrane, or a loaded pre-filter — and WHO sets no health-based TDS guideline, so "TDS 12" is not a safety statement. India's service chain makes these failure modes routine: ~37.7 M waterborne-disease cases a year, AMC-dependent service visits, and documented counterfeit-cartridge incidents.

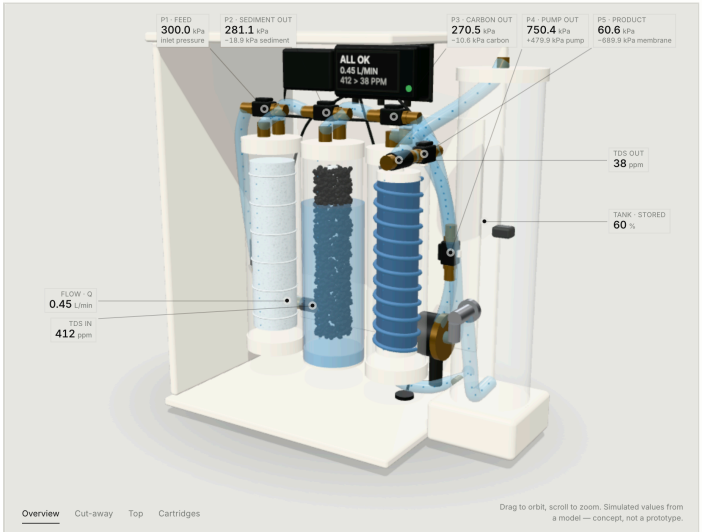
Our solution: a pressure tap on both sides of every element in the water path — including the booster pump. At steady flow, ΔP ÷ Q is a **fingerprint of each cartridge**, recorded when new and re-checked at every service: a loaded stage climbs (≈3× baseline), a removed or bypassed stage collapses (<0.35×), and the pump's measured **boost** gives a motor-health signal no domestic purifier carries today. Any water-path fault **locks the dispense valve** — detection that ends in protection, not just a warning light.

5 + 1 + 2 PRESSURE TAPS · FLOW · TDS PROBES	**0.35×** BASELINE THRESHOLD = CARTRIDGE LEFT THE PATH	**+ΔP/Q** PUMP BOOST WATCHED FOR A WEAK MOTOR	**4** FAIL-SAFES: INTERLOCK, DRY-RUN, LEAK, TANK-FULL

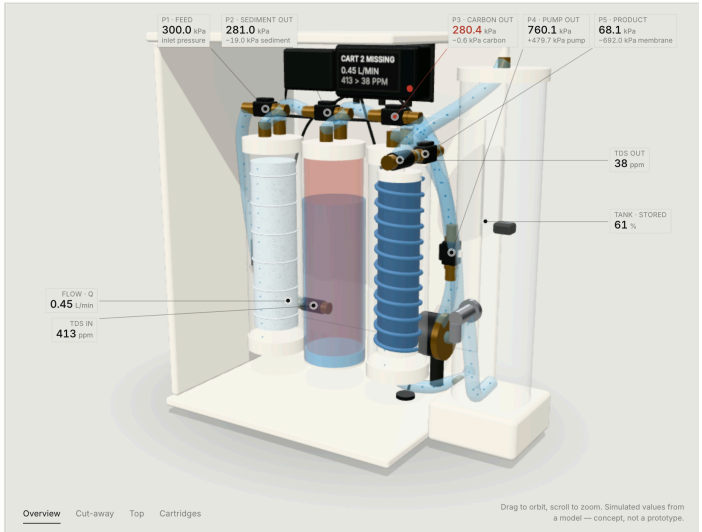
3 COMPONENTS AND MODULES COMPLETED SO FAR

Module	Status and evidence	Sourcing
• Water-path physics model	Per-stage ΔP = k·Q·R with flow normalisation; pump boost with flow droop; verified ladder P1 300 → P2 281 → P3 271 → P4 751 → P5 61 kPa at 0.45 L/min	n/a
• Interactive 3D design twin	Cut-away model of the full water path (pump, tank, faucet, reject line, valves); 9 simulated states; live trace, table and controller display; runs on any phone browser	n/a
• Four-state validation protocol	New / clogged / removed / bypassed + interlock test; bench targets written as targets (±4% repeatability, <0.35× for removed/bypassed)	n/a
• Bill of materials + bench plan	India-priced, vendor-pinned (Robu / KitsGuru / RO-spare markets); ten-day build sequence with cut-list	ordered
◐ Bench rig (hardware)	Components ⟦ordered on 25 Oct / in hand⟧; plumbing and ESP32 firmware per the Week-1 plan in the abstract; first live ΔP logging scheduled 26–28 Oct	in transit
◐ Firmware (ESP32-S3)	Architecture fixed: 2 × ADS1115 (16-bit I2C) for all analogue channels, 20 Hz sampling with pump-cycle averaging, OLED status; bench bring-up this week	in hand

Sourcing risks flagged in the abstract — status: the critical risk was water-safe pressure sensing. Resolved by specification: the common hobby die (XGZP6847) is **gas-only**, so we specified stainless-diaphragm G1/4 transducers (0–1.2 MPa, 0.5–4.5 V, ₹686–1,149 ea, verified in stock at KitsGuru/Robu) read through ADS1115 instead of the ESP32 ADC. All other line items have an Indian supplier with a verified listing; no import dependencies remain.



Normal operation. Five pressure taps, booster pump plumbed between carbon and membrane, product line to the storage tank, reject line to drain. Simulated values — the bench rig exists to replace them with measurements.



The fault today's purifiers cannot see. Carbon housing refitted **without its cartridge**: the stage's normalised drop collapses to 0.06×, the display names "CART 2 MISSING", the dispense interlock locks the tap — while TDS out (38 ppm) is unchanged.

4 AMOUNT SPENT SO FAR AGAINST THE ABSTRACT'S ESTIMATE

Head	Abstract estimate	Committed / spent	Notes
Sensors (5 × pressure, flow, 2 × TDS) + ADS1115 ×2	₹3,010	₹⟦ ⟧	stainless transducers + 16-bit I2C ADC
Pump, valves, float switch, flow restrictor	₹1,824	₹⟦ ⟧	E-Chen 75–100 GPD, 2 × solenoid, FR-450
Controller, OLED, relays, power (SMPS + buck)	₹2,178	₹⟦ ⟧	ESP32-S3, 24 V 3 A SMPS, separate logic rail
Plumbing, cartridges, membrane, tank, enclosure	₹4,150	₹⟦ ⟧	10" cartridges, 75 GPD membrane, 7 L tank
Contingency + reference gauge (validation only)	₹2,038	₹⟦ ⟧	glycerin gauge shared lab equipment
Total prototype budget	**₹13,200**	**₹⟦ ⟧**	**abstract estimated ₹⟦9,000–12,000 — confirm abstract figure⟧**

Fill every ⟦ ⟧ from your order invoices before export. If a head differs from the abstract, keep the abstract column as written and explain the delta in §5 — judges check this table against the video.

5 CHANGES SINCE THE ABSTRACT

- **Track and area of exploration: unchanged.** Same problem, same solution class — measured per-stage filter health.
- **Revised target claim (additions, not pivots):** the abstract's four pressure sensors became **five taps with the pump between stages** — the pump is where a domestic RO actually raises pressure, and the tap pair around it yields a **pump-health signal** at the cost of one sensor. The instrumented path is now feed → sediment → carbon → **booster pump** → membrane → product → storage tank, matching a real domestic RO rather than a laboratory chain.
- **Added a fail-locked dispense interlock** to the safety set (abstract proposed detection only): on any water-path fault the dispense valve refuses to open. Reason: Round 2's brief asks for the status of a safety mechanism — a detection that cannot act was the wrong target claim.
- **Budget revised:** ₹⟦9,000–12,000⟧ → **₹13,200** (pump, solenoids, tank and the fifth transducer). No change to timeline; the ten-day bench plan in the abstract is on schedule — hardware ordering completed 25 Oct, four-state test planned 30 Oct.

6 STATUS OF THE SAFETY / FAIL-SAFE MECHANISMS

Fail-safe	Trigger → action	Status
◐ Fail-locked dispense	any stage < 0.35× baseline → dispense solenoid refuses, "DO NOT USE" on display	logic simulated end-to-end in the design twin (verified, reproducible); firmware task this week; solenoid in hand
◐ Dry-run guard	P1 ≈ 0 kPa (no feed) → pump disabled, auto-resume when water returns	simulated; one firmware comparison on the live P1 channel
◐ Leak shutdown	flow with the tap closed → inlet solenoid closes, pump stops	simulated; floor leak probe is two wires + a pull-up (₹0 effectively)
◐ Tank-full stop	float switch opens at full storage → production pauses, resumes on draw	simulated incl. auto-resume; float switch in hand
• Pump watch	boost < 0.75× its flow-normalised baseline → service warning before flow dies	simulated ("Weak pump" state); comes free with the five-tap layout

Round 3 will demonstrate the interlock and dry-run guard **working on the bench rig**, benchmarked against NSF/ANSI 58's automatic shut-off expectation and IS 16240:2023's recovery requirement.

7 PLANNED NEXT STEPS FOR ROUND 3

- **26–29 Oct:** bench plumbing and leak-test; ESP32 bring-up; zero/span calibration of all five taps against the reference gauge (±2% FS) and a static water column (9.81 kPa/m).
- **30 Oct – 1 Nov:** the four-state test at 0.25/0.40/0.55 L/min ×3 repeats; dust-loaded sediment characterisation; interlock and dry-run guard live in firmware; every reading logged with temperature.
- **1–3 Nov:** measure → twin: replace the 3D model's simulated constants with measured baselines, making it the rig's public dashboard.
- **Round 3 (if shortlisted):** integrate the instrumented path into the purifier shell; 2-minute video of the prototype in operation with the interlock demonstrated; mentor-pairing on manufacturing cost (target: sensor-set BOM < ₹2,000 at volume) and on the pump-health claim, which is Atomberg's home ground.